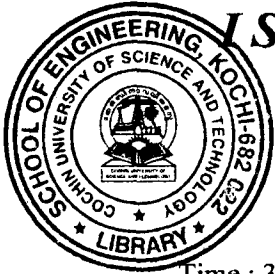


--	--	--	--	--	--	--	--

B.Tech. Degree II Semester Regular and Supplementary/ I Semester Supplementary Examination May 2017



CS/EC/IT GE 15-1205B & CE/EE/ME/SE GE 15-1105A
BASIC MECHANICAL ENGINEERING
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Differentiate between open, closed and isolated system.
- (b) What is meant by compounding of turbines? Explain.
- (c) Compare Otto and Diesel cycles and highlight the differences.
- (d) Explain Co-efficient of Performance of a refrigeration system.
- (e) Discuss specific speed of turbine and how it is employed in the selection of turbine.
- (f) Explain Fuel rods in Nuclear power plant and Pen Stock in Hydro-electric power plant.
- (g) What is Scientific Management? Explain briefly.
- (h) What is meant by depreciation? Explain.
- (i) Explain any two major welding defects.
- (j) What is the function of a clutch? Explain.

PART B

(4 × 10 = 40)

- II. (a) Explain Zeroth law of thermodynamics and temperature scale. (4)
- (b) A vessel having a capacity of 0.85 m³ contains steam at 12 bar and 0.92 dry. Steam is blown off from the vessel until the pressure drops to 6 bars. The valve is then closed and the vessel is cooled until the pressure is 4 bars. Assuming during the blowing off period, enthalpy per kg of steam remains constant in the vessel, determine: (6)
 - (i) the quantity of steam blown off.
 - (ii) the quantity of steam in the vessel after cooling.
 - (iii) net change in entropy.
 - (iv) heat transferred by steam per kg during cooling.

Properties of steam from steam table for the problem is given below.

Absolute Pressure	Temperature	Volume at t_s		Enthalpy			Entropy		
bar	$^{\circ}\text{C}$	m^3/kg		kJ/kg			kJ/kg K		
		Water	Dry Saturated Steam	Water	Vaporization	Steam	Water	Vaporization	Steam
p	t_s	v_f	v_g	h_f	h_{fg}	h_g	s_f	s_{fg}	s_g
12	188	0.00113	0.163	798	1984	2782	2.216	4.307	6.523
6	158.8	0.00110	0.315	670	2085	2755	1.931	4.830	6.761
4	143.6	0.00108	0.462	605	2134	2739	1.776	5.121	6.891

OR

- III. (a) Discuss the function of Steam Traps and Economizer in a boiler. (4)
- (b) Heat is supplied to 1 kg of air at 40 bar and 627°C at constant pressure till the volume is doubled. The air is then expanded isentropically till the pressure drops to 1 bar. Calculate the work done, change in internal energy and heat transfer: (i) at the end of constant pressure heating (ii) at the end of isentropic expansion. Take $R = 287 \text{ J/kgK}$, $C_v = 0.718 \text{ kJ/kgK}$ for air. (6)
- IV. (a) Discuss about carburetor and MPFI system in automobiles (4)
- (b) Derive and determine thermal efficiency and mean effective pressure of Otto cycle with compression ratio of 5 and working between 21 bars and 1 bar respectively as peak and lowest pressure, assuming $\gamma = 1.4$. (6)

OR

- V. (a) Explain summer and winter air conditioning systems. (4)
- (b) Explain the working of a vapour compression refrigeration system with a neat figure. (6)
- VI. (a) Explain the constructional details and working of a Pelton wheel turbine. (4)
- (b) With a neat sketch, explain the various parts and functioning of a Hydro electric power plant. (6)

OR

- VII. (a) Differentiate between centrifugal pumps and reciprocating pumps. (4)
- (b) Explain the time and motion study. (6)
- VIII. (a) With neat sketch, explain (i) extrusion (ii) rolling. (4)
- (b) Derive expression for ratio of belt tensions and power transmitted in belt drives. (6)

OR

- IX. (a) Explain the processes Annealing, Hardening and Normalizing. (6)
- (b) What are the classifications of gears? Explain. (4)